

Training Configuration and Summary:

Sequencing:

- timesteps: 30
- batch_size: 128

Model Hyperparameters:

- min_units: 128
- max_units: 256
- units_step: 64
- dropout_min: 0.2
- dropout_max: 0.3
- dropout_step: 0.1
- activation_functions: ['tanh', 'relu', 'elu']
- optimizers: ['adam', 'rmsprop', 'Nadam']

Tuning:

- max_epochs: 30
- hyperband_iterations: 3

LSTM Multi-Step Training Report

Report Generated: 2025-04-16 13:06:23

Training Start: 2025-04-16_11-26-58

Training End: 2025-04-16_12-45-26

Training Duration: 78.48 minutes

Best Hyperparameters:

Number of Layers: 2

Activation: tanh

Optimizer: rmsprop

Dropout Rate: 0.20

MAE: 0.8942

MSE: 1.4057

RMSE: 1.1856

R2 Score: 0.7677

Multi-Step Forecast:

Forecast Steps: 5

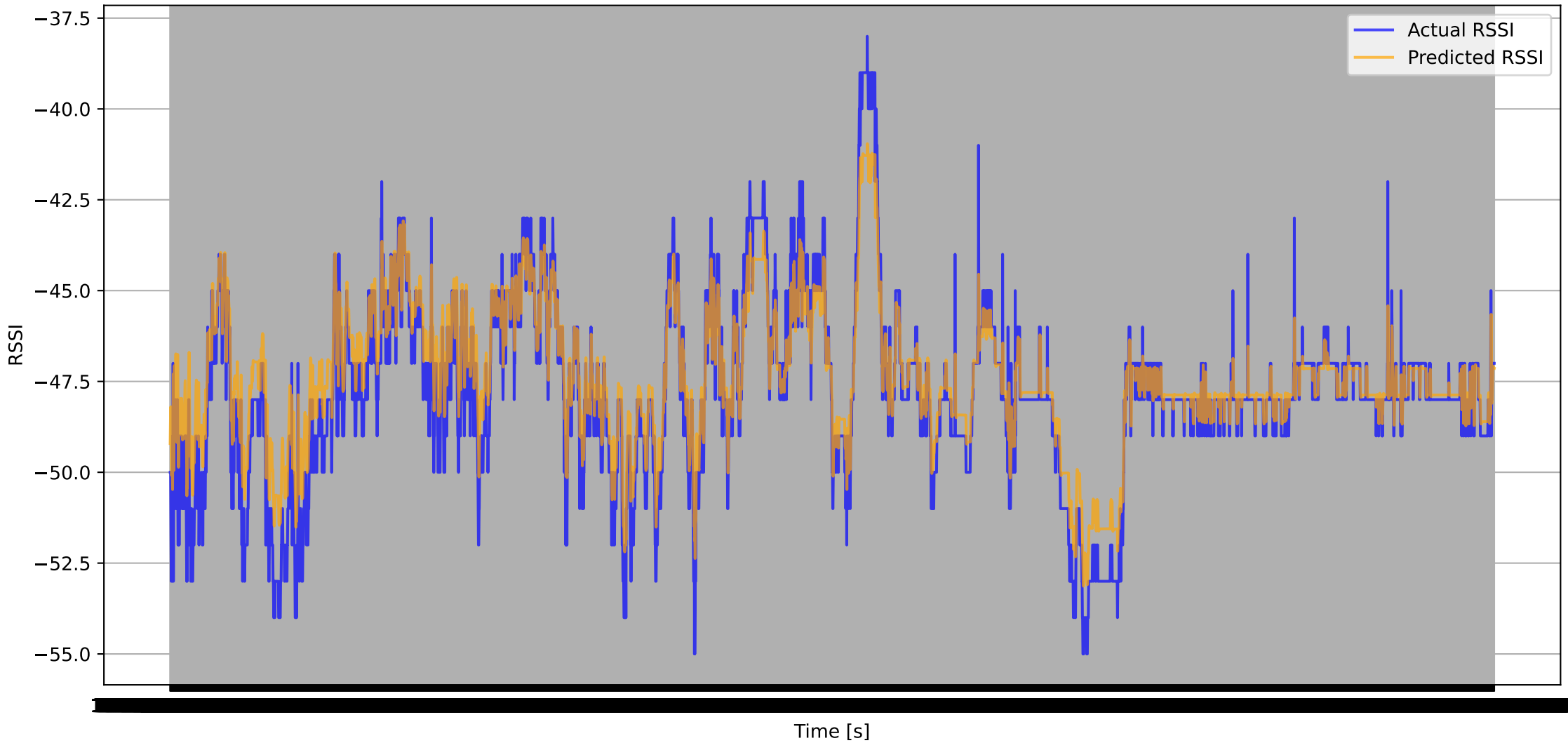
MAE: 0.0256

MSE: 0.0012

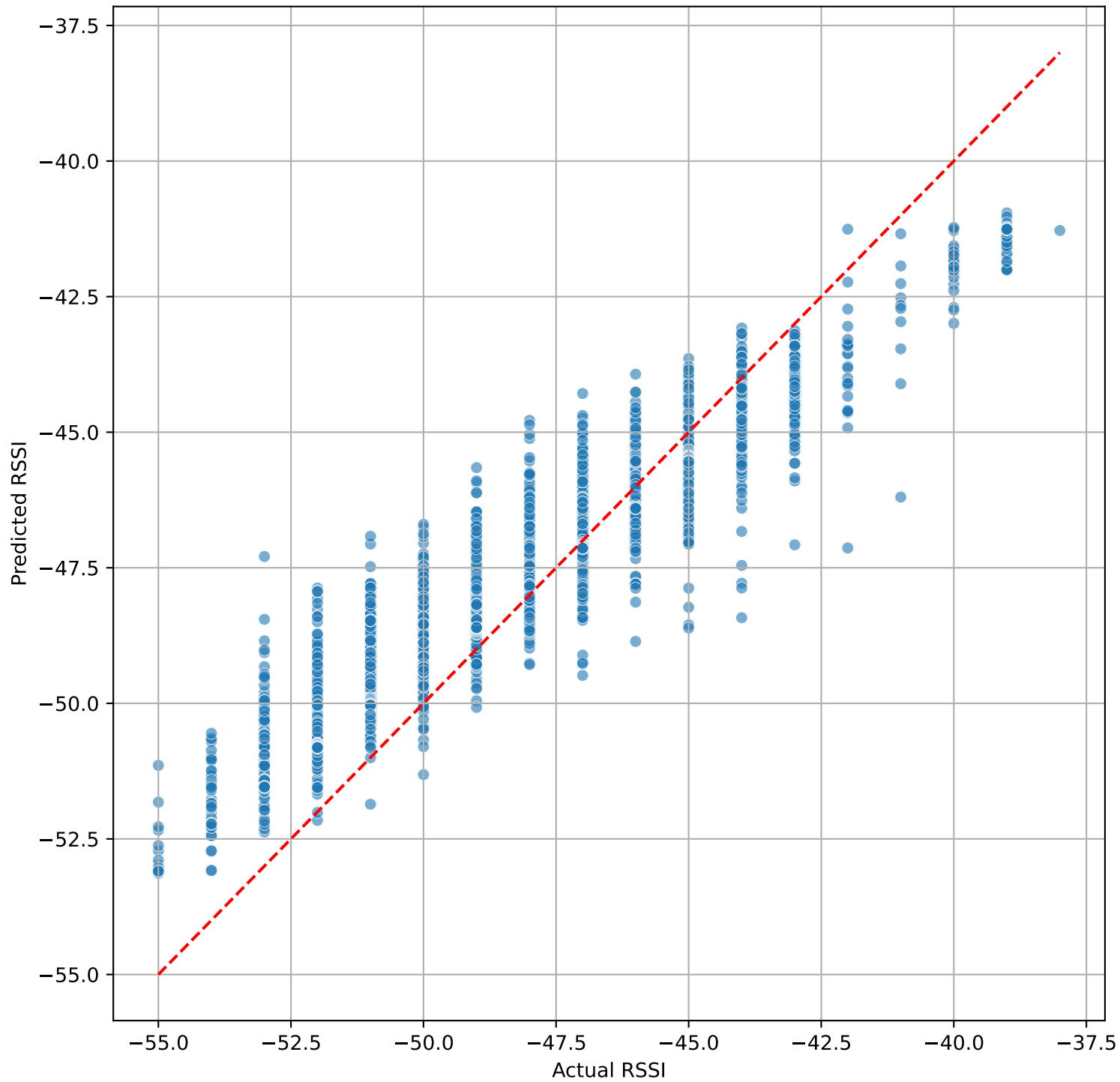
RMSE: 0.0352

R2 Score: 0.8279

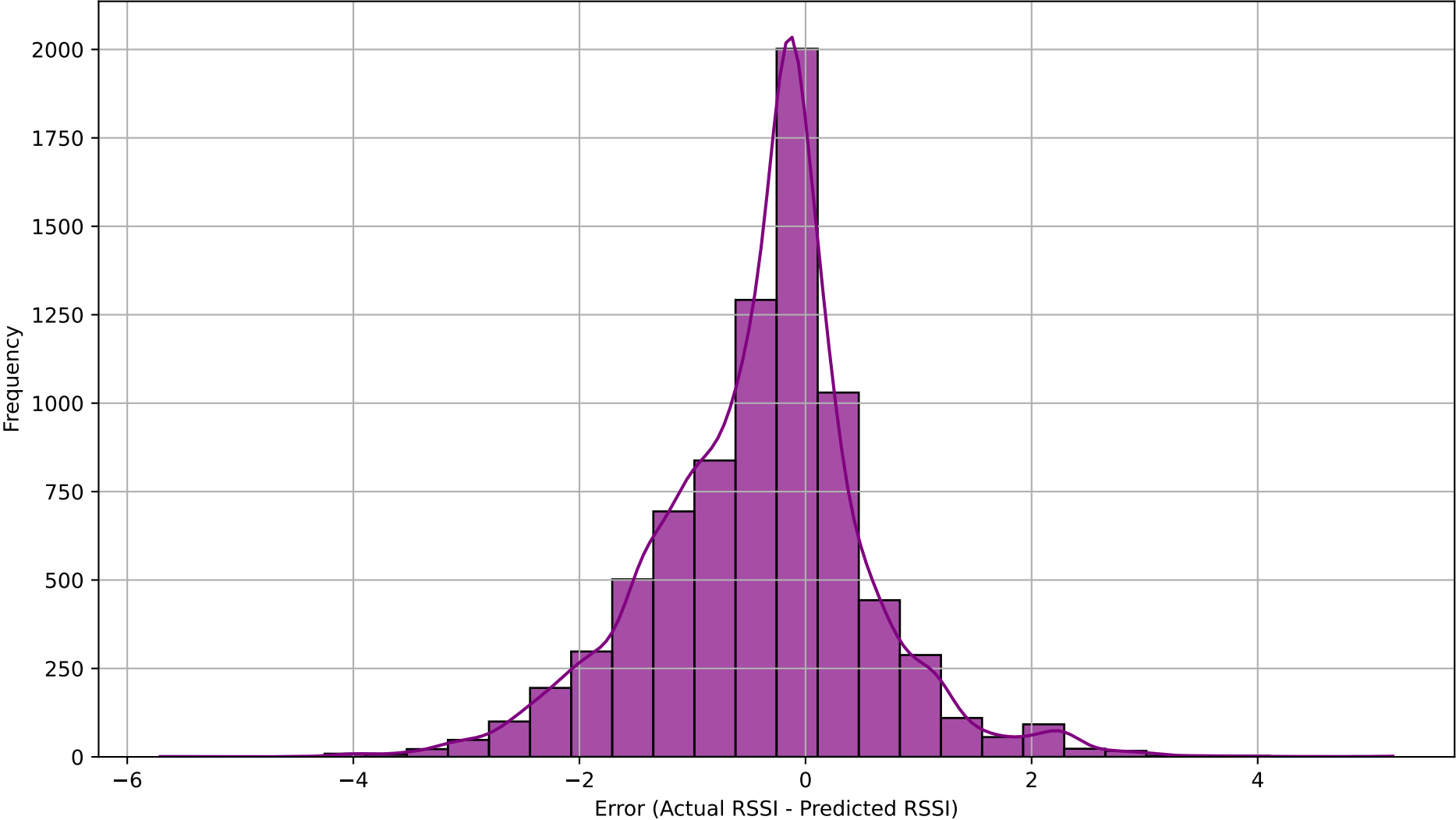
Actual vs Predicted RSSI (First Step)



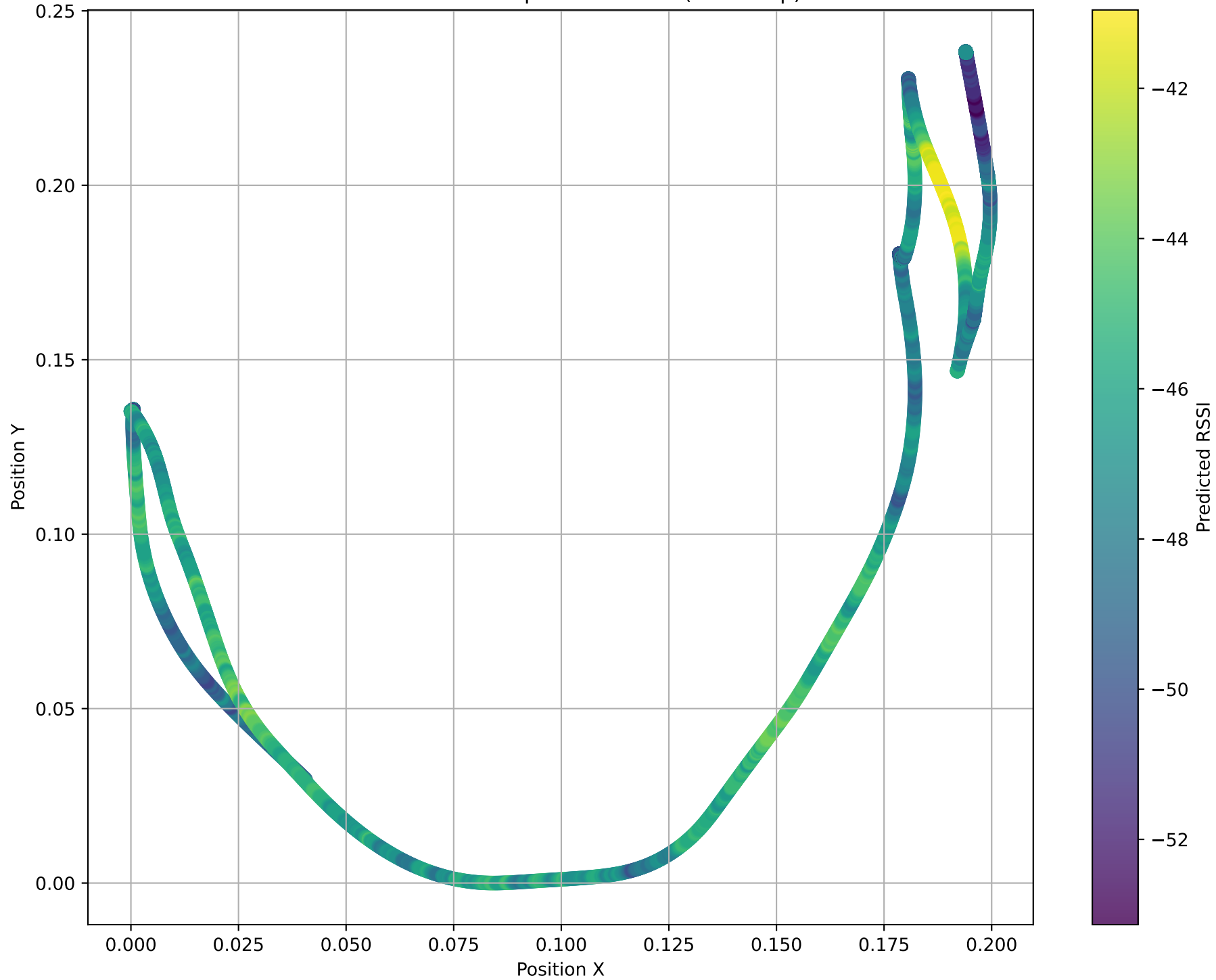
Actual vs Predicted RSSI (First Step)



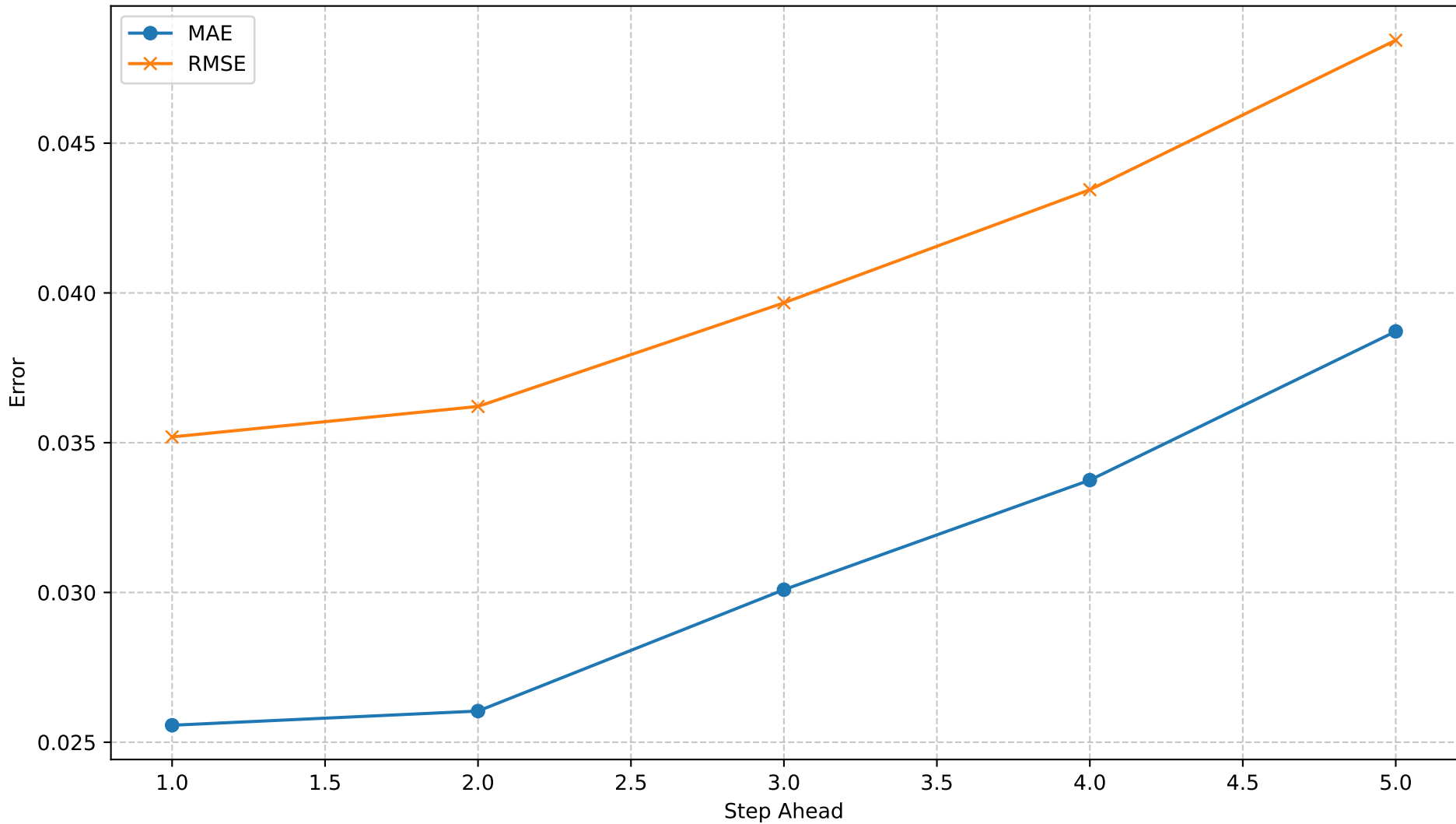
Distribution of Prediction Errors (First Step)



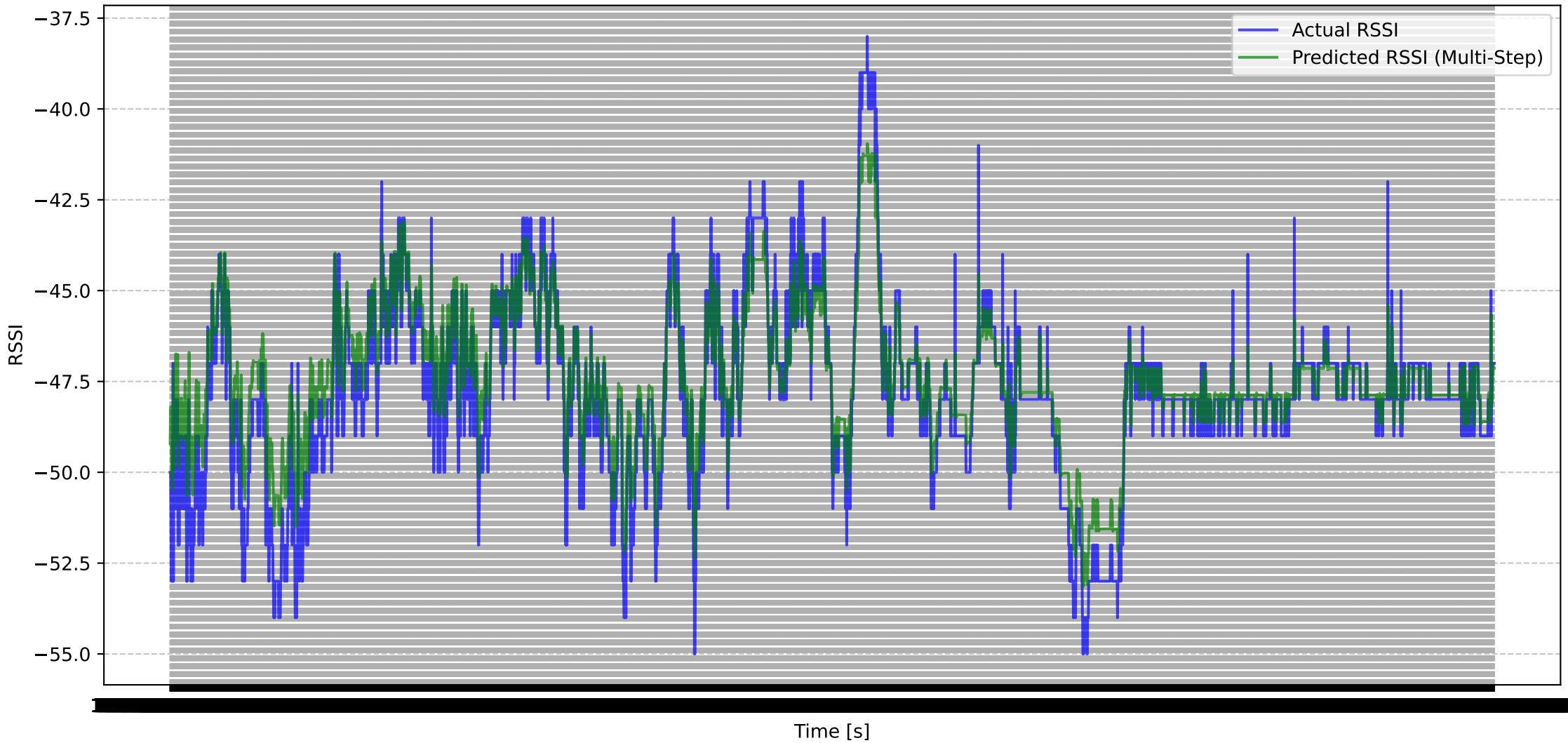
Predicted RSSI in Spatial Context (First Step)



Multi-Step Forecast Error by Step



Actual vs Predicted RSSI (First Step vs Multi-Step)



Resumo Automático:

- Forecast Steps configurados: 5
- Passos com MAE ≤ 2.0 : 5 passo(s)
- MAE no último passo: 0.0387

[V] O modelo mantém boa acurácia durante quase toda a sequência de previsões.